MOSAIC: Data-Certified Stochastic Release under Structured Deployment Shift

Anonymous submission

Abstract

Probe failure on one validation split does not guarantee that an edited release token controls *within-label* source distinguishability for every downstream rule after deployment shift. We present MOSAIC, a software-only method that jointly selects a task-specific stochastic release channel and decoder, or abstains. One simultaneous confidence region over pre-release token laws covers the entire same-table continuum of channels. A new bridge program uses labeled target data to certify the largest retained mass explained by a common Markov transform; arbitrary source-specific contamination is charged to exact channel capacity. We prove exact finite-sample envelopes for the Bayes source attacker within every task label and worst-stratum task error, global solvability for finite alphabets, and impossibility under unrestricted differential shift or a missing target source. In 1,000 hash-locked trials at a safety boundary, naive continuum selection deploys contract-violating releases on 42.1% of tables; MOSAIC records none. Across 100 registered jobs on five real benchmarks, unconditional deployment violates 38/80 estimable external diagnostics; validation-only and bridge plug-ins violate 18/60 and 7/47. MOSAIC deploys 20 BiasBios releases with 0/20 observed violations and abstains on the remaining four benchmarks under the primary contract. Every decision and population risk is retained for replay.

1 Introduction

Representation editing is commonly evaluated by fitting a probe to a frozen embedding (Ravfogel et al. 2020, 2022; Belrose et al. 2023; Jourdan et al. 2023; Avitan, Goldberg, and Elazar 2026), although probe design can change the apparent conclusion (Pimentel et al. 2020). This leaves two deployment gaps: a different downstream rule may recover the attribute, and an edit selected on a noisy validation table may fail after population shift. Both matter when the released token is a public interface.

We study a task-specific finite-token interface. A tokenizer fixed without the certification fold maps an internal representation to $C \in [K]$; a source-blind stochastic channel $M \in \mathrm{Stoch}(K, L)$ releases the only public token Z. Within each task label, the contract bounds the balanced accuracy of the Bayes-optimal source attacker, not a chosen probe. It does not claim to hide source information carried only by label prevalence. Pre-release shift is a common Markov transform plus bounded source-specific contamination, permitting large source-blind drift while charging the part that can create new source differences.

MOSAIC, Minimax-Optimized Source-Agnostic Invariant Channels, builds an exact attacker envelope from simultaneous multinomial confidence regions. Because the event precedes M, it covers every stochastic channel, including one optimized on the same table. A bridge LP uses labeled target data to learn a common transform and the largest certifiable retained mass; exact transform-and-contamination envelopes then certify source distinguishability and a jointly selected decoder. Finite-alphabet optimization is global, and MOSAIC returns ABSTAIN when no feasible release exists.

Randomized channels, confidence regions, contraction, and abstention are not individually new. The contribution is their shift-aware composition: an exact same-table universal-attacker certificate, a data-certified target bridge, matching robust utility, sharp impossibility boundaries, and a globally solved release-or-abstain rule. Hash-locked experiments compare plug-in and held-out selection, LTT, deterministic release, shift-unaware certification, official erasers, and an exact population oracle.

2 Related Work

Erasure and universal representations. Concept-erasure methods remove source-associated directions, covariance, or components (Ravfogel et al. 2020, 2022; Belrose et al. 2023; Jourdan et al. 2023; Avitan, Goldberg, and Elazar 2026); interventions and leakage audits explain why a weak probe is not a universal certificate (Elazar et al. 2021; Chowdhury et al. 2025). FARE certifies every downstream classifier for restricted finite encoders, while Fair Normalizing Flows bounds strong adversaries through tractable densities (Jovanovic et al. 2023; Balunović, Ruoss, and Vechev 2022). MOSAIC instead places its confidence event on pre-release token laws, uniformly covering a stochastic post-channel optimized on that table and transferred through an explicit shift class.

Randomized mappings and risk control. Probabilistic preprocessing and fair representation already show the value

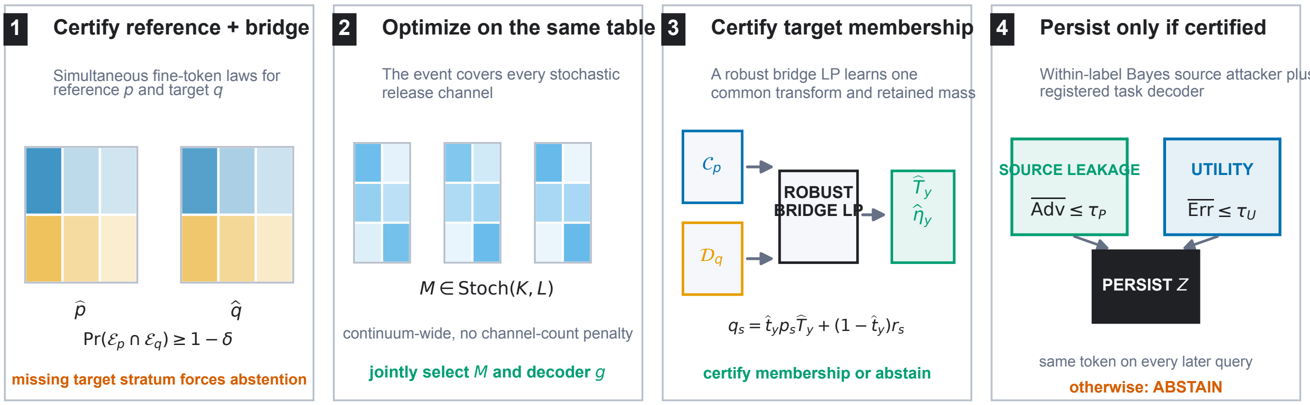

Figure 1: MOSAIC jointly certifies reference and labeled target-bridge token laws, optimizes a continuum of source-blind stochastic channels on that event, and uses a robust bridge LP to certify target membership. The runtime samples one persistent token only when within-label source distinguishability and registered task utility pass; otherwise it abstains. The primary contract is within-label source distinguishability.

of stochastic channels (Calmon et al. 2017; Makhdoumi et al. 2014; Agarwal and Deshpande 2024). Risk-controlling prediction, Learn-Then-Test (LTT), conformal risk control, and Pareto testing certify registered decisions and may abstain (Bates et al. 2021; Angelopoulos et al. 2025, 2024; Laufer-Goldshtein et al. 2023). Robust variants address specified covariate or uncertainty shifts (Tibshirani et al. 2019; Cauchois et al. 2024). These are the closest decision-layer precedents. MOSAIC instead covers an uncountable channel family through one confidence region for its sufficient table, without testing channels separately.

Shift, comparison of experiments, and abstention. Dobrushin coefficients quantify channel contraction (Diaz, Sankar, and Kairouz 2018); fairness and domain-adaptation impossibility results show why unrestricted shift precludes universal guarantees (Lechner et al. 2021; Agarwal et al. 2025; David et al. 2010). WILDS and subgroup-robustness benchmarks document the practical problem (Koh et al. 2021; Sagawa et al. 2020; Zhou et al. 2021; Sohoni et al. 2020). Bounded-shift fairness transfer and distributional fairness certification give complementary predictor-level robust bounds (Chen et al. 2022; Kang et al. 2022). Common source-blind transformation is Blackwell garbling, while Le Cam deficiency handles approximate experiment comparison (Blackwell 1953; Le Cam 1964; Torgersen 1991); arbitrary residuals follow contamination neighborhoods (Huber 1964). Testable learning asks when target data can validate a shift assumption; recent work studies rejection granularity and tests registered noise models (Klivans, Stavropoulos, and Vasilyan 2024; Chandrasekaran et al. 2025; Patel et al. 2026; Goel et al. 2026). MOSAIC contributes a finite-confidence garbling-and-contamination bridge that composes with the same-table optimizer. Its contract-driven abstention is distinct from predictive-confidence rejection (El-Yaniv and Wiener 2010; Geifman and El-Yaniv 2017).

3 Problem Setup

Let $S \in [G]$ denote source, $Y \in [J]$ task label, and $C \in [K]$ the fine token. For a fixed label, write $p_{s,y}(c) = \Pr(C = c \mid S = s, Y = y)$. The source-blind release channel M produces $Z \in [L]$. The balanced accuracy of the strongest downstream source attacker within label y is

$$A_y(P, M) = \frac{1}{G} \sum_{z=1}^{L} \max_{s \in [G]} (p_{s,y} M)(z), \qquad (1)$$

with normalized advantage $\mathrm{Adv}_G(A) = (GA - 1)/(G - 1)$. This is chance-normalized and ranges from zero to one. The registered source contract is $\max_y \mathrm{Adv}_G(A_y) \leq \tau_P$.

This target must not be read as unconditional source privacy. To certify an attacker on the natural mixture, define the joint fine symbol $X = (Y, C)$, estimate $P(X \mid S)$ without balancing labels, and lift the channel as $\widetilde{M}((y, c), z) = M(c, z)$. The same envelope below then certifies source inference from Z; using output (Y, Z) certifies the stronger attacker also given the true label. Our experiments use the registered within-label target, so they do not estimate this prevalence channel.

For each supported source–label stratum, the certification table supplies an empirical law $\widehat{p}_{s,y}$ and radius $\epsilon_{s,y}$. The simultaneous confidence set is $\mathcal{C}_{s,y} = \{p \in \Delta_K : \|p - \widehat{p}_{s,y}\|_1 \leq \epsilon_{s,y}\}$. Under conditionally i.i.d. categorical sampling within each registered source–label stratum, the simultaneous event

$$\mathcal{E} = \bigcap_{s,y} \{\|\widehat{p}_{s,y} - p_{s,y}\|_1 \leq \epsilon_{s,y}\} \qquad (2)$$

has probability at least $1 - \delta$ after a fixed allocation across strata. We use the Weissman multinomial radius (Weissman et al. 2003) in the confirmation, although any valid simultaneous region can replace it. The tokenizer and fine alphabet must be fixed without these observations.

The external law for label y belongs to the structured class

$$q_s = t_y p_s T_y + (1 - t_y) r_s, \qquad t_y \geq 1 - \eta_y, \qquad (3)$$

Method	Finite Z	Bayes attacker	Same-table M	Target shift	Shift evidence
FARE (Jovanovic et al. 2023)	yes	yes	no	no	no
Randomized fair reps. (Agarwal and Deshpande 2024)	yes	yes	population	no	no
LTT / risk control (Angelopoulos et al. 2025; Bates et al. 2021)	general	candidate	finite family	optional	model-specific
Blackwell / deficiency (Blackwell 1953; Le Cam 1964)	experiment	all decisions	no	population	assumed
Robust validation (Cauchois et al. 2024)	no	no	no	declared ball	estimated
Testable shift (Klivans, Stavropoulos, and Vasilyan 2024)	no	no	hypothesis	yes	tested
MOSAIC	yes	exact	yes	pre-release	finite-sample

Table 1: Component-level scope of the closest frameworks. "Same-table continuum" means that one confidence event remains valid after optimizing an uncountable stochastic-channel family on that table. No individual column is a novelty claim; MOSAIC's contribution is the complete final row.

where T_y is a source-independent fine-token channel from a registered convex uncertainty set and each residual r_s is arbitrary. Both components pass through the same release channel M. This placement is crucial: a constant-row software channel cannot leak information that appears before it.

4 Adaptive Within-Label Attacker Certificate

For a bounded vector v, define the exact confidence-set support function

$$\Phi(\widehat{p}, \epsilon; v) = \max_{p \in \Delta_K : \|p - \widehat{p}\|_1 \le \epsilon} p^\top v. \tag{4}$$

It is obtained by moving at most $\epsilon/2$ mass from low to high coordinates of v, or from an equivalent linear-program dual. For attacker assignment $a \in [G]^L$, let $v_{M,a,s}(c) = \sum_z M(c,z)\mathbf{1}\{a(z) = s\}$. We then define

$$\overline{A}(\widehat{P}, M) = \frac{1}{G} \max_{a \in [G]^L} \sum_s \Phi(\widehat{p}_s, \epsilon_s; v_{M,a,s}). \tag{5}$$

Theorem 1 (Adaptive exact envelope)**.** *On $\mathcal{E}$, simultaneously for every row-stochastic M, including a channel selected as an arbitrary function of the same table, $A(P, M) \le \overline{A}(\widehat{P}, M)$. For fixed M, the right-hand side is the exact supremum over the product of the stated ℓ_1 confidence sets.*

The proof writes the Bayes attacker as a maximum over deterministic assignments. For a fixed assignment the source confidence sets separate, giving one support function per source; all maxima are finite and commute. The implication is pointwise over the complete stochastic-channel space, which is why adaptive selection needs no channel-count correction.

5 Data-Certified Shift Membership

The structured model in Eq. (3) is useful only when membership is evidenced rather than presumed. Let a labeled bridge sample from the target population define simultaneous confidence sets $\mathcal{D}_{s,y}$ for its fine-token laws $q_{s,y}$. Write

$$h_{s,y}(w) = \max_{p \in \mathcal{C}_{s,y}} p^\top w, \qquad \underline{q}_{s,y}(z) = \min_{q \in \mathcal{D}_{s,y}} q(z). \tag{6}$$

For each label, the bridge program chooses retained mass t and $W = tT$ by solving

$$\begin{aligned} \max_{t,W} \quad & t \\ \text{s.t.} \quad & W \ge 0, \quad W\mathbf{1} = t\mathbf{1}, \quad 0 \le t \le 1, \\ & h_{s,y}(W_{:z}) \le \underline{q}_{s,y}(z) \quad \forall s, z. \end{aligned} \tag{7}$$

The support functions and coordinate lower bounds have exact LP duals, so Eq. (7) is one linear program rather than a transform search.

Theorem 2 (Adaptive finite-sample bridge)**.** *Suppose the reference and bridge confidence sets jointly cover every $p_{s,y}$ and $q_{s,y}$. Any feasible $(\widehat{t}, \widehat{W})$ with $\widehat{t} > 0$, including the same-table maximizer, defines $\widehat{T} = \widehat{W}/\widehat{t}$ and residual laws satisfying*

$$q_{s,y} = \widehat{t}\, p_{s,y} \widehat{T} + (1 - \widehat{t}) r_{s,y} \quad \forall s. \tag{8}$$

Moreover, Eq. (7) *returns the largest retained mass certified uniformly over the product confidence region.*

Indeed, feasibility and coverage imply $q_{s,y}(z) \ge p_{s,y}^\top \widehat{W}_{:z}$ coordinatewise. The difference is nonnegative and sums to $1 - \widehat{t}$, so normalization constructs $r_{s,y}$. Conversely, robust membership requires exactly these coordinatewise inequalities; maximizing the reference support and minimizing the bridge coordinate yields Eq. (7). Thus the learned singleton transform $\{\widehat{T}_y\}$ and contamination $\widehat{\eta}_y = 1 - \widehat{t}_y$ can be inserted into the certificates below on the same event. If a required bridge source–label stratum is absent, its confidence set is the full simplex, every coordinate lower bound is zero, and Eq. (7) forces $\widehat{t} = 0$.

6 Certification under Pre-Release Shift

We use three names consistently: the *data-certified bridge* tests shift membership, the *transform-exact certificate* enumerates the learned shift geometry, and the *capacity fallback* upper-bounds it when that geometry cannot be enumerated.

Define the arbitrary differential-shift capacity

$$\mathrm{Cap}_G(M) = \frac{1}{G} \max_{a \in [G]^L} \sum_s \max_c v_{M,a,s}(c). \tag{9}$$

This equals $\sup_{r_1,\ldots,r_G} A(R, M)$. For $G = 2$, its normalized value is the Dobrushin coefficient, the largest total variation distance between two rows of M.

For a transform polytope with registered extremes $\mathcal{V}_y = \mathrm{ext}(\mathcal{T}_y)$, define the shifted envelope

$$\begin{aligned} \overline{A}_y^X(M) = \frac{1}{G} \max_{T \in \mathcal{V}_y,\, a \in [G]^L} \sum_s \Big[& \\ & (1 - \eta_y)\Phi(\widehat{p}_{s,y}, \epsilon_{s,y}; T v_{M,a,s}) \\ & + \eta_y \max_c v_{M,a,s}(c) \Big]. \end{aligned} \tag{10}$$

Theorem 3 (Transform-exact structured-shift certificate). *On $\mathcal{E}$, every same-table selected M and every external law in Eq.* (3) *satisfy $A_y(Q, M) \le \overline{A}_y^X(M)$. For fixed M, the right-hand side is the exact supremum over the stated confidence sets, transform polytope, retained masses, and arbitrary residual laws.*

For fixed T, retained mass, and attacker assignment, the source confidence sets and residual simplexes separate. Their support values are respectively $\Phi(\widehat{p}, \epsilon; Tv)$ and $\max_c v(c)$. The latter is no smaller, so the worst retained mass is $1 - \eta_y$. The support objective is convex in T; its maximum over a polytope occurs at an extreme. These finite maxima commute, proving exactness. Since the statement holds pointwise for every M, it survives adaptive channel selection.

When transform extremes cannot be enumerated, a capacity-fallback corollary is available. Define $\rho_T(M) = \max_c \mathrm{TV}((TM)(c, \cdot), M(c, \cdot))$ and $B_y(M) = \min\{\mathrm{Cap}_G(M), \overline{A}_y(\widehat{P}, M) + \rho_{\mathcal{T}_y}(M)\}$. Then

$$A(Q, M) \le (1 - \eta_y) B_y(M) + \eta_y \, \mathrm{Cap}_G(M). \tag{11}$$

The exact envelope in Eq. (10) is never larger. When $\eta_y = 0$ and $TM = M$, both reduce to the reference certificate; when $\eta_y = 1$, both reduce to exact distribution-free capacity.

Because the fallback separately upper-bounds every law in the same structured class, both transform-exact source and utility values are pointwise no larger than their fallback values. "Never worse" is therefore an implementation check implied by the theorem; experiments measure how often tightness changes the release decision. Proofs are in the supplement.

For decoder $g : [L] \to [J]$, let $u_y(c) = \sum_z M(c, z) \mathbf{1}\{g(z) \ne y\}$. The matching exact conditional error certificate is

$$\begin{aligned}\overline{E}_{s,y}^X(M, g) = \max_{T \in \mathcal{V}_y} \Big[&(1 - \eta_y) \\ &\Phi(\widehat{p}_{s,y}, \epsilon_{s,y}; Tu_y) + \eta_y \max_c u_y(c) \Big].\end{aligned} \tag{12}$$

It is the exact supremum over the same confidence and shift sets. Thus M, g, within-label source distinguishability, and worst-stratum utility can all be selected on one event.

Theorem 4 (No free lunch). *$\mathrm{Cap}_G(M) = 1/G$ if and only if all rows of M are identical. For two sources, if two fine tokens carry different task labels and incur row-wise decoder errors at most e_0 and e_1, then the normalized capacity is at least $1 - e_0 - e_1$.*

Consequently, unrestricted source-specific shift permits exact erasure only by making the public token independent of the fine token. A bounded η_y is therefore an identifiable assumption of the deployment contract, not a hidden technical convenience.

There is a separate identifiability boundary for external auditing. Suppose a required source is absent and let $A_{\mathrm{miss}}(q_0, M) = \sup_{q_1 \in \Delta_K} A((q_0, q_1), M)$.

Theorem 5 (Missing-source audit). *For a release channel M fixed independently of the missing-source law and audit randomness, any protocol with false-certification probability at most δ uniformly over the unobserved law q_1 can claim $A((q_0, q_1), M) \le \tau < A_{\mathrm{miss}}(q_0, M)$ with probability at most δ.*

The observed audit distribution is identical for every missing law; choosing a simplex maximizer makes every stronger claim a false certification. The same argument makes the utility bound for a missing source–label stratum at least $\max_c(M\ell_y)(c)$. Missing required strata are therefore unestimable, not evidence of safety; the supplement gives the full proof.

7 Global Optimization and Abstention

For fixed decoder, support functions, transformed scores, and residual maxima have linear epigraphs. Enforcing all transform extremes and G^L attacker assignments gives one LP in the KL channel entries; enumerating J^L decoders yields the global finite-alphabet optimum. The capacity fallback uses one mixed-integer LP per decoder. Acceptance requires solver optimality, valid primal residuals, and agreement with separately recomputed certificates.

Release protocol. Fix tokenizers, alphabets, thresholds, split roles, and confidence allocation before viewing certification or bridge folds. Construct simultaneous regions, solve Eq. (7), and enumerate decoder LPs subject to every source constraint. Replay each returned certificate, then select the feasible registered tokenizer with minimum certified error under a fixed tie break. Zero retained mass, missing support, no feasible row, or a failed numerical check returns ABSTAIN; otherwise only (M, g) and Z are released.

With $R = \max_y |\mathcal{V}_y|$, the direct formulation is polynomial in K for fixed (G, J, L, R) but exponential in the deliberately small L through attacker and decoder enumeration. Larger public alphabets require separation or column generation. Endpoint enumeration, population identities, dense channel grids, and randomized exact-versus-fallback comparisons check the encoding; the supplement gives complexity and power-curve details.

8 Hash-Locked Evaluation

Synthetic confirmation. Before outcomes, we SHA-256 locked the protocol, code, evaluators, seed exclusions, gates, and theory curve. Two source–label populations, two shift regimes, eight rules, and up to five sample sizes use 1,000 independent tables per cell and familywise $\delta = .05$. A separate population evaluator grades every selected channel, and replay scripts reconstruct every decision, risk, interval, and optimizer check. The primary contrasts test unsafe plug-in selection and safe retention against held-out certification, LTT, and deterministic release; every registered cell remains reportable.

Real bridge confirmation. We registered 100 fresh jobs: 20 untouched seeds on each of Waterbirds, Camelyon17-WILDS, CivilComments-WILDS, BiasBios-Clinical, and GaitPDB, spanning vision, medical vision, text, and time series. Each job evaluates 13 frozen feature edits from the official INLP, LEACE, R-LACE, TaCo, and MANCE++ implementations; proxies are forbidden. Construction data alone fit a balanced task score and its four-bin tokenizer. Separate reference observations certify the pre-shift table, while the natural external split is divided within each supported source–label stratum into two-thirds labeled bridge data and an untouched one-third diagnostic fold. Neither diagnostics nor their outcomes enter channel or candidate selection.

The bridge labels are a deliberate data requirement, not metadata inferred by the method: it needs target S and Y to estimate $q_{s,y}$ and test the registered within-label contract. Task labels alone could support retraining, but cannot establish source-conditional token laws, audit every downstream source rule, or certify that a target-trained model obeys this release contract. Where S is sensitive, its collection, consent, and permitted audit use are deployment prerequisites; without a supported stratum MOSAIC abstains.

The familywise level is .05 over 13 candidate tables and both reference and bridge regions. The source-advantage contract is .35; worst-stratum utility thresholds are $.30, .35, .40, .45$, and $.49$, with $.40$ primary. We globally optimize $L = 2$, then reoptimize $L = 4$ only for the best covered $L = 2$ candidate. A strict replay accepts no decision tolerance and an exact-rational audit checks every serialized bridge slack and outward risk bound. After the first strict audit exposed an overconservative structural-zero implementation error, we locked the one-condition correction before rerunning all 100 jobs; both versions remain public. Locked comparators use the same candidates and folds: capacity transfer, a bridge plug-in without intervals, a validation plug-in that ignores the bridge, and unconditional deployment of the best validation plug-in. All datasets, thresholds, abstentions, and missing-support outcomes remain reportable.

Bridge validity stress. A separate 12,000-table confirmation varies the bridge sample size from 500 to 5,000 per source–label stratum. One population obeys a common transform; two violate the registered model through either source-specific transforms or contamination above the declared budget. Acceptance requires a finite-sample retained-mass lower bound of at least .80. This study tests whether valid membership gains power with data while these concrete violations remain rejected; it is not a test against every possible misspecification.

9 Audited Results

The preregistered synthetic gates pass. Figure 2 reports every implementable rule in the primary cells; "capacity fallback" is MOSAIC's generic capacity certificate. Independent replay recomputes all 64,000 method receipts, 128,000 population risks, decisions, and aggregate intervals with zero mismatch.

The paired safety difference is 42.1 points (95% bootstrap interval $[39.0, 45.2]$), with 421 plug-in-only violations and no reverse discordance. Across 8,000 capacity-fallback certification tables, no accepted release violates the exact population contract. In the retention cell, capacity-fallback safe deployment exceeds held-out, finite-LTT, and deterministic rates by 53.8, 53.0, and 56.3 points. No deterministic channel is feasible. These finite-run facts support, but do not replace, the $1 - \delta$ guarantee.

A separately locked 5,000-table refinement evaluates exact shift geometry on the same tables. At $n = 125$, transform-exact certification safely deploys on 53.0% versus 0.3% for fallback; at $n = 250$, the rates are 99.9% and 57.8%. The exact objective is no larger on all 5,000 pairs, as theory requires, and all 10,000 method-cell decisions have zero false acceptances. Full cell tables, threshold sensitivity, amendment history, and audits are in the supplement.

Matched finite-family baselines. A separately locked replay places four rules on the same 1,000 tables in each primary scenario. In the retention cell, continuum MOSAIC certifies 579/1,000 deployments (57.9%, exact 95% interval $[54.77, 60.98]$), versus 283/1,000 (28.3%, $[25.53, 31.20]$) for Holm LTT over all 500 fixed channel–decoder pairs. MOSAIC alone deploys on 312 paired tables, LTT alone on 16, and both on 267, a 29.6-point paired difference. The exact McNemar value is 2.28×10^{-72}; it is a post-audit descriptive statistic, not a preregistered gate. The confidence-region grid and matched deterministic restricted-encoder adaptation both abstain throughout this cell. All four rules have zero observed false acceptance. The deterministic row tests FARE's finite-encoder principle under MOSAIC's shift contract, not FARE's official end-to-end training objective.

Bridge validity and power. Across a separate 12,000-table locked stress study, a correctly specified common transform is accepted on 0%, 31%, 100%, and 100% of tables as the bridge stratum size rises through 500, 1,000, 2,000, and 5,000. Underdeclared contamination has population retained mass $.775 < .80$; a source-specific transform has minimum retained mass .50. Both violating populations are rejected on every table at every sample size. Independent replay finds no failure on the joint confidence event and no invalid false acceptance. This supports power against the two registered violations, not detection of arbitrary misspecification.

Real data-certified bridge. At the primary error contract $.40$, strict MOSAIC deploys on 20/100 jobs (20.0%, exact 95% interval $[12.67, 29.18]$), all externally estimable, with 0/20 observed violations (one-sided 95% upper bound 13.91%). The selected official edits comprise six INLP, five LEACE, and nine MANCE++ candidates. Capacity transfer deploys none. The bridge plug-in deploys 47 and violates 7/47 estimable diagnostics (14.9%); validation-only selection deploys 80, including 20 unestimable Camelyon jobs, and violates 18/60 estimable diagnostics (30.0%). Unconditional deployment violates 38/80 estimable jobs (47.5%) and also releases all 20 unestimable jobs. MOSAIC retains 20 safe deployments at every stronger threshold $.30$–$.45$ and adds three safe Waterbirds deployments at $.49$; the supplement enumerates every threshold. These untouched diagnostics expose concrete selection failures but do not replace the finite-sample bridge guarantee or establish clinical safety.

Table 2 is intentionally less flattering than the aggregate: the primary strict result has releases on one of five benchmarks and abstains on the other four. The 100 jobs are repeated seeds, not 100 independent deployment populations. A descriptive dataset-resampling interval for the strict deployment rate is $[0, .60]$; leaving out BiasBios leaves

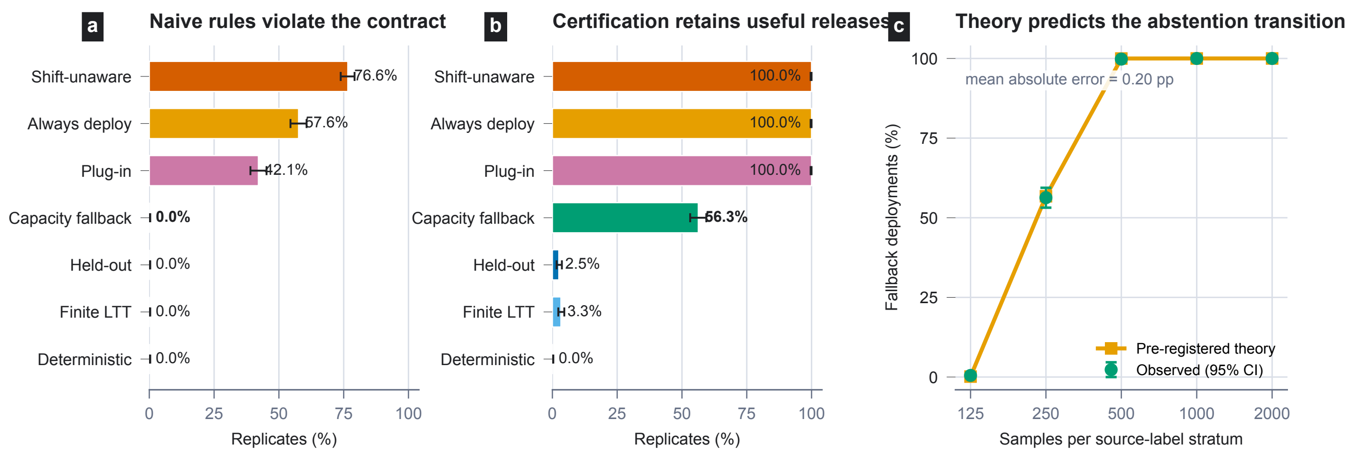

Figure 2: Hash-locked confirmation with 1,000 independent certification tables per cell. Error bars are exact 95% Clopper–Pearson intervals. (a) At the safety boundary, plug-in continuum selection falsely accepts 42.1% of tables; MOSAIC abstains with no false acceptance. Ignoring shift is worse. (b) In the retention cell, the capacity fallback safely deploys on 56.3%, versus 2.5% for held-out certification, 3.3% for finite LTT, and 0% for deterministic MOSAIC. (c) Deployment follows the preregistered theory curve with 0.20 percentage-point mean absolute error. Exact 95% Clopper–Pearson intervals are shown.

Dataset	Strict	Bridge	Validation	Unconditional	Strict diagnostic	Strict limitation
BiasBios-Clinical	20/0	20/0	20/0	20/0	20 estimable	median $t = .797$
Camelyon17-WILDS	0/0	0/0	20/–	20/–	none	missing support
CivilComments-WILDS	0/0	0/0	0/0	20/20	none	no joint feasibility
GaitPDB	0/0	7/7	20/18	20/18	none	missing support
Waterbirds	0/0	20/0	20/0	20/0	none	no joint feasibility

Table 2: Dataset-grouped primary results, 20 seeds per benchmark. Entries are deployments/diagnostic violations among estimable deployments; "–" denotes an unestimable diagnostic, never evidence of safety. The 20 strict MOSAIC deployments all occur on BiasBios-Clinical.

zero strict deployments, whereas leaving out any other dataset leaves 20. Thus the job-level interval summarizes the registered trials only and does not support a cross-domain deployment-rate claim. Waterbirds makes the safety–retention tradeoff especially clear: its bridge plug-in releases 20/20 jobs with no observed diagnostic violations, while strict MOSAIC abstains because it cannot certify joint feasibility. That diagnostic cannot establish the formal contract. Missing-support abstentions are required by the theorem, but they are not observed safety successes.

What the real ablations remove. The primary rows form a diagnostic ladder rather than unrelated baselines. Capacity transfer keeps simultaneous regions but replaces the learned shift geometry by a worst-case channel-capacity charge; its 0/100 deployments versus MOSAIC's 20/100 show that the exact bridge is decision-relevant. The bridge plug-in retains target data and the fitted transform but removes confidence regions, gaining 27 deployments at the cost of seven observed violations. Validation-only selection removes the target-membership gate, adding 60 deployments but producing 18 estimable violations and 20 unsupported releases. Unconditional deployment additionally removes contract-based abstention and produces 38 violations. Thus intervals, bridge membership, exact geometry, and abstention each change a measured deployment outcome.

Corrective numerical audit. The original raw rule produced the same 20 primary deployments, but its tolerance-qualified inequalities could not support an exact software claim. A pre-inspection strict v1 replay then overcorrected: it replaced an exactly zero transform-output denominator by the positive numerical guard, making a vacuous constraint force retained mass to zero and yielding 0/100 deployments. After all v1 outcomes were visible, we hash-locked one structural correction before any v2 rerun. A scope audit compares 2,600 label certificates: every one of the 918 changed labels, and no unchanged-condition label, contains the required exactly zero output column. All original inequalities are still rechecked. Deterministic replay matches 100/100 v2 receipts, and exact rational arithmetic verifies all 1,300 bridge slacks and 1,400 outward release bounds. We report v2 as post-audit corrective evidence and preserve raw, v1, and v2 receipts; it is not presented as an untouched preregistered outcome. A separate pre-outcome lock on new seeds 1220–1224 then confirms the corrected pipeline: it deploys on all 5 BiasBios jobs with 0/5 diagnostic violations and abstains on the remaining 20. Independent replay covers 325 candidates and 350 global optimizations; exact-rational replay passes every serialized certificate. This confirms the correction, not a broader cross-dataset deployment rate.

Alphabet sensitivity. Reoptimizing the best covered candidate with four release tokens is no worse than its two-token solution on all 100 jobs within the registered 10^{-7} solver criterion (largest serialized difference 1.07×10^{-8}), but is strictly better on only five. The real result therefore comes mainly from certifying shift geometry, not from hiding utility in a larger alphabet. It also reinforces the intended scope: MOSAIC is a compact task-specific release interface, not a generally reusable edited embedding.

10 Discussion

What changes relative to candidate testing. LTT and Pareto testing control risk after registering a finite family. MOSAIC instead certifies the categorical experiment that precedes release: on one covered table, every M and g inherit the same pointwise envelope. The optimizer can therefore search a continuum without converting its internal points into separate hypotheses. This is not free adaptivity. A tokenizer or eraser fitted separately changes the sufficient table and receives its own multiplicity allocation. The matched 500-pair replay isolates this distinction: even a strong structure-aware Holm implementation discards power that remains available when the whole channel space is downstream of one event.

What the bridge establishes. The bridge is a finite-sample membership certificate, not an assertion that all deployment shift is benign. It asks how much of every target source law can be dominated by one common transformation of the corresponding reference law. The unmatched remainder is charged adversarially. More bridge data can tighten this one-sided statement; unsupported strata force zero retained mass. A source-specific transform can sometimes mimic a common transform at the token level, so no finite test detects every violation. The guarantee is precisely for laws admitted by the certified categorical bridge, and later population change requires a new bridge.

How the shift set is obtained. The real protocol does not ask a user to guess a smoothing transform or tune a contamination level until a favorable release appears. The user registers the source and task strata, tokenizer, target sampling population, confidence allocation, and decision thresholds. Equation (7) then ranges over every source-blind $K \times K$ Markov transform and returns the largest simultaneously supportable retained mass t_y; the certificate itself sets $\eta_y = 1 - t_y$. Thus a large common change is permitted when target data support it, while every unexplained coordinate is charged adversarially. Registering a different target population, source definition, or tokenizer is a new statistical object and requires a new bridge rather than reuse of the old certificate.

Why a bridge rather than target retraining. Retraining on labeled target task data can improve predictive performance, and is an appropriate alternative when the desired output is simply a new target predictor. MOSAIC answers a narrower release problem: whether one persistent, source-blind token interface obeys a pre-registered within-label source and utility contract. Its bridge therefore requires target source–label audit data, while a task-only retraining procedure need not. The methods are complementary rather than substitutes; target retraining does not by itself certify the stated source contract.

Operational meaning. MOSAIC deliberately releases less than a general embedding. It trades arbitrary future task reuse for a task-specific token whose complete source-inference experiment can be enumerated and persisted. This is most appropriate when software exposes a stable decision interface and abstention is acceptable, such as a fixed triage or screening stage. It is not a claim of cryptographic secrecy, causal fairness, or safety of unregistered metadata. The practical output is consequently auditable: a channel, a decoder, a versioned persistence rule, and either a contract certificate or ABSTAIN.

11 Guarantee Semantics

The probability $1 - \delta$ is over simultaneous reference and bridge regions. On that event, the conclusion is uniform over optimized channels and every law in the certified shift class; it does not assume that a validation interval covers an external metric. A frozen tokenizer's table covers the channel continuum and every decoder without a channel-count correction. Separately learned tokenizers or erasers have different tables and therefore receive a registered familywise allocation. MOSAIC pays for statistical objects, not every point optimized inside one covered object.

Threat surface. The attacker knows M and may apply any rule to Z. The source-blind software channel is the only public token interface and samples once per immutable item; persistent repeats reveal no more than one Z. If r fresh draws are permitted, certification must use product channel $M^{\otimes r}$; unbounded draws can reveal its row. Transcripts over distinct items require a joint-law certificate. Releasing C, original features, source-dependent metadata, mutable identifiers, or another side channel is outside scope.

12 Limitations and Scope

The bridge requires conditionally i.i.d. reference and target observations, including source and task labels, for every required source–label stratum and does not cover later drift. These audit labels may be sensitive and require a lawful, consented collection and governance plan. Tokenizers are fixed independently of certification; large alphabets may need structured regions or column generation. MOSAIC trades reusable embeddings for a certifiable task interface and durable persistence state. It certifies within-label source inference from one released item, not label-prevalence leakage, person-level privacy across multiple items, separate interfaces, every downstream harm, or utility forbidden by the contract.

13 Conclusion

MOSAIC treats editing as release design. A pre-channel confidence event permits continuum optimization; a target bridge separates common drift from differential contamination. Impossibility results identify required abstention, while locked experiments show unsafe refusals and safe retention under certified shift geometry.

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